Mar. 26 Chapter 10: Group Homomorphisms

- **Def** If $(G,*)$ & $(H,\star)$ are groups, a function $\varphi: G \to H$ such that $\varphi(a*b) = \varphi(a) \star \varphi(b)$ $\forall a,b \in G$ is called a homomorphism

- **Def** The kernel (null space in linear alg.) of a homomorphism $\varphi: G \to H$ is the set $\ker\varphi = \{a \in G : \varphi(a) = e_H\}$ "All the stuff that gets mapped to identity)

- **Def** The image of $\varphi: G \to H$ is the set $\varphi(G) = \{h \in H : h = \varphi(g) \text{ for some } g \in G\}$

Ex $\rho: \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ with $\rho(a) = \bar{a}$

$\rho(a+b) = \overline{a+b} = \bar{a} + \bar{b}$. Thus ρ is a homomorphism.

Image $\rho(\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$, $\ker\rho = \{x \in \mathbb{Z} : n \mid x\} = n\mathbb{Z}$

Ex $GL_2(\mathbb{R}) = \{\text{invertible } 2\times 2 \text{ matrices with coefficients in } \mathbb{R}\}$

$\det: GL_2(\mathbb{R}) \to \mathbb{R}^\times$ $\quad \det(A \cdot B) = \det(A) \cdot \det(B)$, thus det is a homomorphism

$\det\left(\begin{bmatrix} 1 & 0 \\ 0 & \pi \end{bmatrix}\right) = \pi$, works for any element of $\mathbb{R}^\times$, so $\det(GL_2(\mathbb{R})) = \mathbb{R}^\times$

$\ker(\det) = \{A \in GL_2(\mathbb{R}) : \det(A) = 1\}$, many matrices, e.g $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} \pi & 0 \\ 0 & 1/\pi \end{bmatrix}$, etc

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- Later: we are going to compare $G/\ker\varphi$ to a subgroup of H.

  More on Thm. 10.1 in book

- <u>Thm. 10.2</u> Let $\varphi: G \to H$ be a homomorphism and let $K \leq G$.
  1) $\varphi(K) \leq H$ Proving this is good practice for exam
  2) If K is cyclic, $\varphi(K)$ is cyclic
  3) If K is Abelian, then $\varphi(K)$ is Abelian
  4) If $K \trianglelefteq G$, then $\varphi(K) \trianglelefteq \varphi(G)$.

  <u>Proof sketch:</u>

  Suppose $K \trianglelefteq G$. We want $\varphi(K) \trianglelefteq \varphi(G)$, so let $\varphi(a) \in \varphi(G)$ be arbitrary. We want $\varphi(a)\varphi(k)\varphi(a)^{-1}$ to be a subset of $\varphi(K)$. Since φ is a homomorphism, $\varphi(a)\varphi(k)\varphi(a)^{-1} = \varphi(aka^{-1}) = \varphi(k')$ for some $k' \in K$. Thus $\varphi(a)\varphi(k)\varphi(a)^{-1} \in \varphi(K)$. Hence $\varphi(K) \trianglelefteq \varphi(G)$.

  little k

  Let $\varphi(a)\varphi(k)\varphi(a)^{-1}$ be an element of

  since $K \trianglelefteq G$, $aka^{-1} \in K$

  Points 5-9 of thm. 10.2 to be discussed later

- <u>Thm. 10.3</u> (The first Isomorphism Theorem) If $\varphi: G \to H$ is a homomorphism, then the mapping $G/\ker\varphi \longrightarrow \varphi(G)$ given by $a\ker\varphi \longmapsto \varphi(a)$ is an isomorphism. Symbolically, $G/\ker\varphi \cong \varphi(G)$.

  FIT

  G mod ker φ, group of cosets

  image of G under φ

  G mod the kernel is isomorphic to the image of G

<u>Ex.</u> $\rho: \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$

$G = \mathbb{Z}$, $\ker\rho = n\mathbb{Z}$, $\mathbb{Z}/\ker\rho \cong \rho(\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$

$\mathbb{Z}/\ker\rho = \mathbb{Z}/n\mathbb{Z}$

<u>Ex.</u> $GL_2(\mathbb{R})/SL_2(\mathbb{R}) \cong \mathbb{R}^\times$

$\det(A) = r$ & $\det(B) = r$

$AA^{-1} = I = BB^{-1} \Rightarrow AA^{-1} = BB^{-1} \Rightarrow AA^{-1}B = B$

$\det(A^{-1}B) = \det(A)^{-1}\det(B) = r^{-1}r = 1$

A and B differ by 1

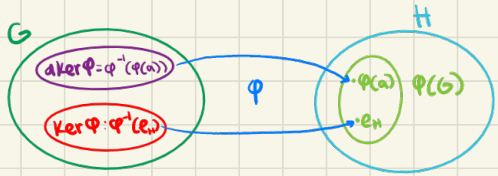
<u>Ex.</u> $\xi: \mathbb{Z} \to \{i, -i, 1, -1\}$ by $n \longmapsto i^n$ (C_4)

$a+b \longmapsto i^{a+b} = i^a i^b$

$\xi(\mathbb{Z}) = \mathrm{im}(\xi) = \{i, -i, 1, -1\}$, $\ker\xi = 4\mathbb{Z}$

FIT: $\mathbb{Z}/\ker\mathbb{Z} \cong \xi(\mathbb{Z}) = C_4$

$\mathbb{Z}/\ker\mathbb{Z} = \mathbb{Z}/4\mathbb{Z}$



- $\varphi^{-1}(h) = \{a \in G : \varphi(a) = h\}$ is the <u>preimage</u> of h

• <u>Proof of 1st Thm:</u> for convenience, define $\overline{\varphi}: G/\ker\varphi \to \varphi(G)$ with $\overline{\varphi}(a\ker\varphi) = \varphi(a)$.

To see $\overline{\varphi}$ is well-defined, suppose $a\ker\varphi = a'\ker\varphi$. We want $\underbrace{\overline{\varphi}(a\ker\varphi) = \varphi(a) = \overline{\varphi}(a'\ker\varphi) = \varphi(a')}$

Since $a\ker\varphi = a'\ker\varphi$, $a\overset{\in a}{e_G} \in a'\ker\varphi$. Thus $a = a'k$ for some $k \in \ker\varphi$.

Hence $\varphi(a) = \varphi(a'k) = \varphi(a')\varphi(k) = \varphi(a')e_H = \varphi(a')$. So $\overline{\varphi}$ is well-defined.

put in cosets w/different representatives but are same coset, $\overline{\varphi}$ will output same thing

showing $\overline{\varphi}$ is a homomorphism/isomorphism to be continued